

NAMJUN CHO

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Education

Case Western Reserve University

Doctor of Philosophy in Biomedical Engineering

Aug. 2025 – May 2031

Cleveland, Ohio

University of Illinois Urbana Champaign

Bachelor of Science in Bioengineering

Aug. 2019 – May 2025

Champaign, Illinois

- Engineering Visionary Scholarship; Illinois Engineering Achievement Scholarship

Relevant Coursework

- Immunoengineering
- Stem Cell Engineering
- Quantitative Pharmacology
- Cancer Nanotechnology
- Tissue Engineering
- Bio Disease Modeling

Experience

Wolstein Research Laboratory - Zhang Lab

August 2025 – May 2031

PhD Candidate

Cleveland, Ohio

- Investigating the impact of the *Malassezia* species on the neurological and neuro-immune system by dissecting the unique metabolisms of *Malassezia* and neuronal cells.
- Establishing various protocols in the lab related to the genetic transformation for *Malassezia*, chemical derivatization, and mass spectrometry analysis. I compare numerous methodologies, materials, and strategies and choose the best or most reasonable method through extensive research and empirical testing.
- Supporting benchtop and clinical studies for targeting CD11b epitopes to promote monocyte-dendritic differentiation using a plant-based PAMP, BG34-200. I employ good manufacturing practices (GMP) to prepare end-user BG34-200 infusion bags, as well as investigate the molecular interaction between BG34-200 and the CD11b epitope.

Carl R. Woese Institute for Genomic Biology - Hudson Lab

June 2023 – August 2024

Research Assistant

Urbana, Illinois

- Spearheaded protoplast isolation and transfection experiments on *Maize*, *Sorghum*, and *Nicotiana benthamiana*, utilizing hemocytometry for precise protoplast quantification and fluorescence microscopy for gene expression analysis.
- Enhanced competency in basic laboratory skills such as lipid/genome/plasmid extraction, PCR, gel electrophoresis, PEG transfection, etc., through regular practice and refinement. Performed almost 100 protocols in just one summer and produced the highest concentration of plasmid seen in the lab group at 1100 ng/μL without sacrificing purity.
- Co-authored an upcoming research publication, contributing extensively to the introduction and discussion sections. Employed tools like Excel, MATLAB, OriginLab, and BioRender for data analysis and visualization, focusing on plant protoplasts, isolation and transfection methods, and various plant species.

Everitt Laboratory - Jang Lab

June 2024 – May 2025

Research Assistant

Urbana, Illinois

- Cultured and maintained neuronal cells for up to 20 days, adhering to protocols established by PI Dr. Jang, ensuring consistent growth, contamination-free environments, and optimal cell viability for subsequent experiments.
- Executed advanced experiments including AAV purification, calcium imaging, optogenetics, and utilized confocal laser scanning microscopy to analyze neuronal behavior.
- Conducted and presented literature reviews to enhance experimental protocols and stay informed on current trends. Taught neuronal cell culture techniques to lab members while learning advanced laboratory methods from colleagues, fostering a collaborative and innovative research environment.

Extracurricular

Tracy Undergraduate Team Science Training Program (TTP)

September 2024 – May 2025

Carl R. Woese Institute for Genomic Biology

Champaign, Illinois

- Collaborated with interdisciplinary teams on 2 case studies in health, agriculture, and energy, enhancing critical research, collaboration, and problem-solving skills.
- Participated in 10 specialized workshops and industry panels, building professional networks with leaders from Eli Lilly, ADM, and Mars-Wrigley, while strengthening communication across scientific disciplines.
- Pursued a Team Science Training Certificate by completing rigorous program activities focused on fostering innovation, team-based research, and cross-industry collaboration.

Skills & Interests

Technical: Matlab, Python, Origin Lab, BioRender, C++, VS Code, Java, Excel

Laboratory: BG34-200 Synthesis, Calcium Imaging, Confocal Microscopy, Fluorescence Microscopy, Fungal Cell Culture, Fungal Engineering, Genome Extraction, Lipid/Genome/Plasmid Extraction, Mass Spectrometry, Nanopore Sequencing, Neuronal Cell Culture, PEG Transfection, Protoplast Isolation